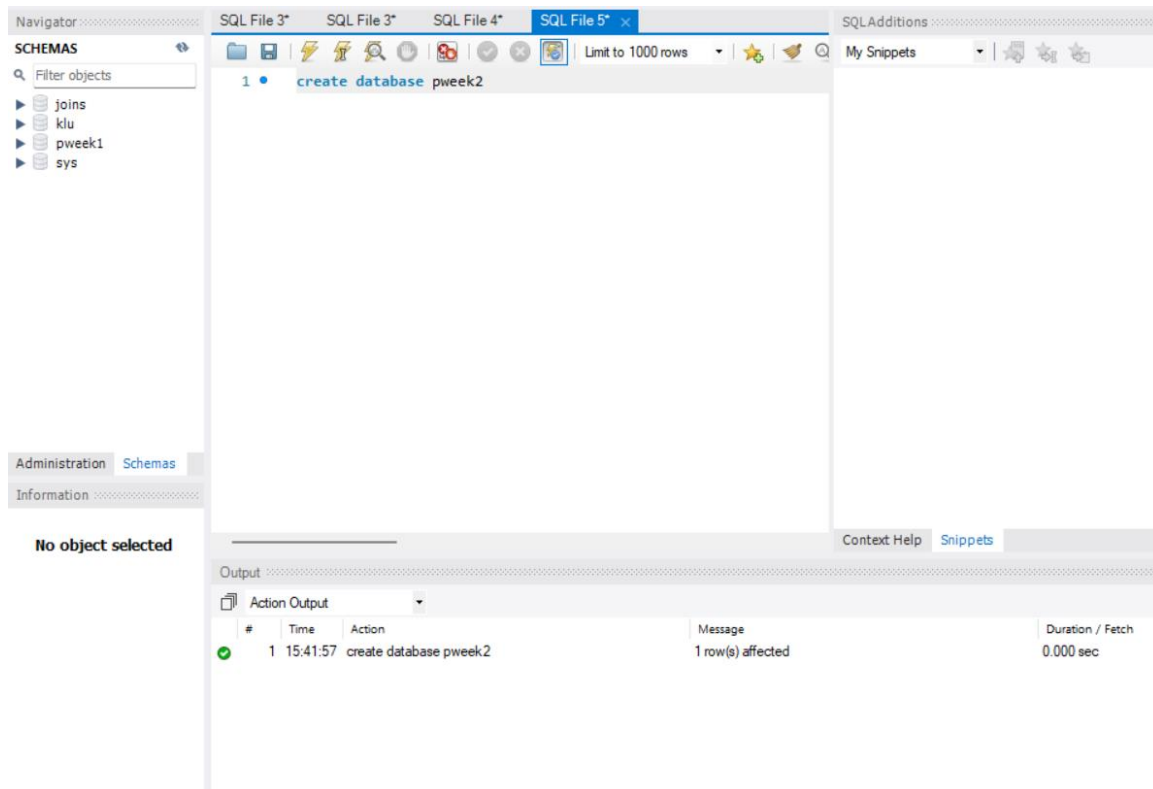
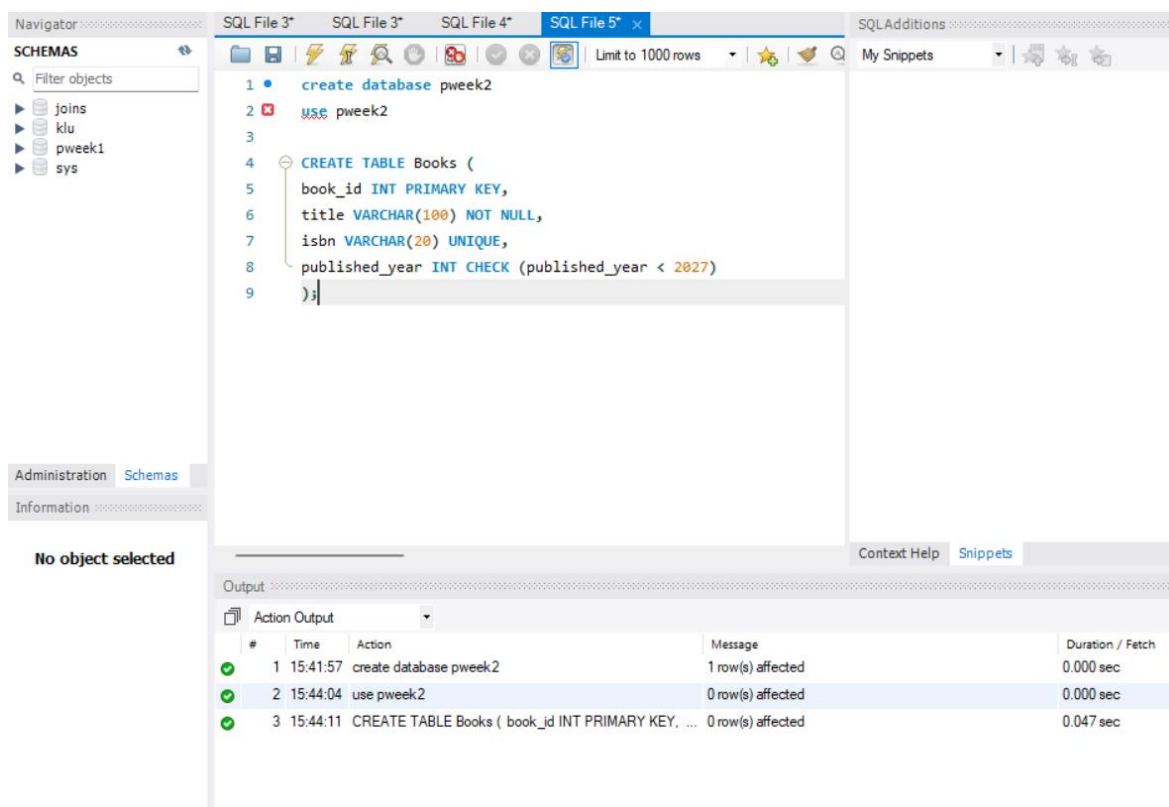


Practical week-2

Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



Step 3 & 4 — Insert Books and Display

Navigator: SQL File 3* SQL File 3* SQL File 4* SQL File 5* x SQLAdditions

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY, ...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_yea...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec

Navigator: SQL File 3* SQL File 3* SQL File 4* SQL File 5* x SQLAdditions

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY, ...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_yea...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	NULL	NULL	NULL

Books 1 x Apply Revert

Step 5, 6 & 7 — Members Table: Create, Insert, Display

The screenshot displays a SQL IDE interface with the following components:

- Navigator:** Shows a tree view of schemas including joins, klu, pweek1, and sys.
- SQL Editor:** Contains the following SQL code:


```

6  title VARCHAR(100) NOT NULL,
7  isbn VARCHAR(20) UNIQUE,
8  published_year INT CHECK (published_year < 2027)
9  );
10
11 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
12 (1, 'The Great Gatsby', '9780743273565', 1925),
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15
16 • SELECT * FROM Books;
17
18 • CREATE TABLE Members (
19     member_id INT PRIMARY KEY,
20     full_name VARCHAR(100),
21     email VARCHAR(100) UNIQUE
22 );
23

```
- Output:** Displays the execution results of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY,...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_ye...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	15:47:19	CREATE TABLE Members (member_id INT PRIMARY...	0 row(s) affected	0.047 sec

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

SQL File 5* x

```

12 (1, 'The Great Gatsby', '9780743273565', 1925),
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15
16 • SELECT * FROM Books;
17
18 • CREATE TABLE Members (
19     member_id INT PRIMARY KEY,
20     full_name VARCHAR(100),
21     email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;

```

SQLAdditions

My Snippets

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
✓ 2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
✓ 3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY,...	0 row(s) affected	0.047 sec
✓ 4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_ye...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
✓ 5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMARY...	0 row(s) affected	0.047 sec
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email) ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec

Object Info Session

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

SQL File 5* x

```

20     full_name VARCHAR(100),
21     email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;

```

SQLAdditions

My Snippets

Context Help Snippets

Result Grid

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com
* NULL	NULL	NULL

Members 2 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
✓ 3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KE...	0 row(s) affected	0.047 sec
✓ 4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
✓ 5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 8 — Create Loans Table (FOREIGN KEYS)

The screenshot shows a SQL IDE interface with a Navigator on the left, a central SQL editor, and an Output pane at the bottom.

Navigator: Shows a tree view with folders for 'joins', 'klu', 'pweek1', and 'sys'. The 'Schemas' tab is active.

SQL Editor: Contains the following SQL code:

```
21 email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;
30
31 • CREATE TABLE Loans (
32 loan_id INT PRIMARY KEY,
33 member_id INT,
34 book_id INT,
35 loan_date DATE,
36 FOREIGN KEY (member_id) REFERENCES Members(member_id),
37 FOREIGN KEY (book_id) REFERENCES Books(book_id)
38 );
```

Output: The 'Action Output' tab is selected, showing a log of SQL actions:

#	Time	Action	Message	Duration / Fetch
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec

Step 9 & 10 — Insert 10 Loans and Display

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x
Limit to 1000 rows

```

43 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
44 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
45 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
46
47 • SELECT * FROM Loans;
48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby

Administration Schemas
Information

No object selected

Result 4 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec

Step 12 — GROUP BY Query

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x
Limit to 1000 rows

```

48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;
53
54 • SELECT published_year, COUNT(book_id) AS Total_Books
55 FROM Books
56 GROUP BY published_year
57 ORDER BY published_year;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

published_year	Total_Books
1925	1
1949	1
1960	1

Administration Schemas
Information

No object selected

Result 5 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec

Step 13–16 — Donation via TRANSACTION (Atomicity)

Navigator

SQL File 3* SQL File 3* SQL File 4* SQL File 5*

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

```

48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;
53
54 • SELECT published_year, COUNT(book_id) AS Total_Books
55 FROM Books
56 GROUP BY published_year
57 ORDER BY published_year;
58
59 • CREATE TABLE Donation_History (
60     donation_id INT PRIMARY KEY,
61     book_id INT,
62     donor_name VARCHAR(100),
63     donation_date DATE,
64     FOREIGN KEY (book_id) REFERENCES Books(book_id)
65 );

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
✓ 10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
✓ 11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec

Object Info Session

SQL File 3*

SQL File 3*

SQL File 4*

SQL File 5* x

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek1

sys

Administration

Schemas

Information

No object selected

52

INNER JOIN Books b ON l.book_id = b.book_id;

53

54

SELECT published_year, COUNT(book_id) AS Total_Books

55

FROM Books

56

GROUP BY published_year

57

ORDER BY published_year;

58

59

CREATE TABLE Donation_History (

60

donation_id INT PRIMARY KEY,

61

book_id INT,

62

donor_name VARCHAR(100),

63

donation_date DATE,

64

FOREIGN KEY (book_id) REFERENCES Books(book_id)

65

);

66

67

START TRANSACTION;

68

69

INSERT INTO Books (book_id, title, isbn, published_year)

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec
15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec

SQL File 3*

SQL File 3*

SQL File 4*

SQL File 5* x

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek1

sys

Administration

Schemas

Information

No object selected

58

59

CREATE TABLE Donation_History (

60

donation_id INT PRIMARY KEY,

61

book_id INT,

62

donor_name VARCHAR(100),

63

donation_date DATE,

64

FOREIGN KEY (book_id) REFERENCES Books(book_id)

65

);

66

67

START TRANSACTION;

68

69

INSERT INTO Books (book_id, title, isbn, published_year)

70

VALUES (4, 'Animal Farm', '9780451526342', 1945);

71

72

INSERT INTO Donation_History (donation_id, book_id, donor_name, donat

73

VALUES (1, 4, 'Raj Kumar', CURDATE());

74

75

COMMIT;

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec
15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec
16	16:25:53	INSERT INTO Books (book_id, title, isbn, published_y...	1 row(s) affected	0.000 sec
17	16:25:59	INSERT INTO Donation_History (donation_id, book_id, ...	1 row(s) affected	0.000 sec
18	16:26:02	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

Navigation

- Schemas
 - joins
 - klu
 - pweek1
 - sys

SQL Editor

```

-- SQL File 3*
-- SQL File 3*
-- SQL File 4*
-- SQL File 5* x
Limit to 1000 rows

70 VALUES (4, 'Animal Farm', '9780451526342', 1945);
71
72 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donat
73 VALUES (1, 4, 'Raj Kumar', CURDATE());
74
75 • COMMIT;
76
77 • CREATE INDEX idx_books_isbn ON Books(isbn);
78
79 • SELECT * FROM Books WHERE isbn = '9780451524935';

```

Result Grid

#	book_id	title	isbn	published_year
▶ 3	1984	9780451524935	1949	
*	HOLL	HOLL	HOLL	HOLL

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec
✓ 15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 16	16:25:53	INSERT INTO Books (book_id, title, isbn, published_y...	1 row(s) affected	0.000 sec
✓ 17	16:25:59	INSERT INTO Donation_History (donation_id, book_j...	1 row(s) affected	0.000 sec
✓ 18	16:26:02	COMMIT	0 row(s) affected	0.000 sec
✓ 19	16:26:43	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
✓ 20	16:26:51	SELECT * FROM Books WHERE isbn = '9780451524...	1 row(s) returned	0.000 sec / 0.000 sec